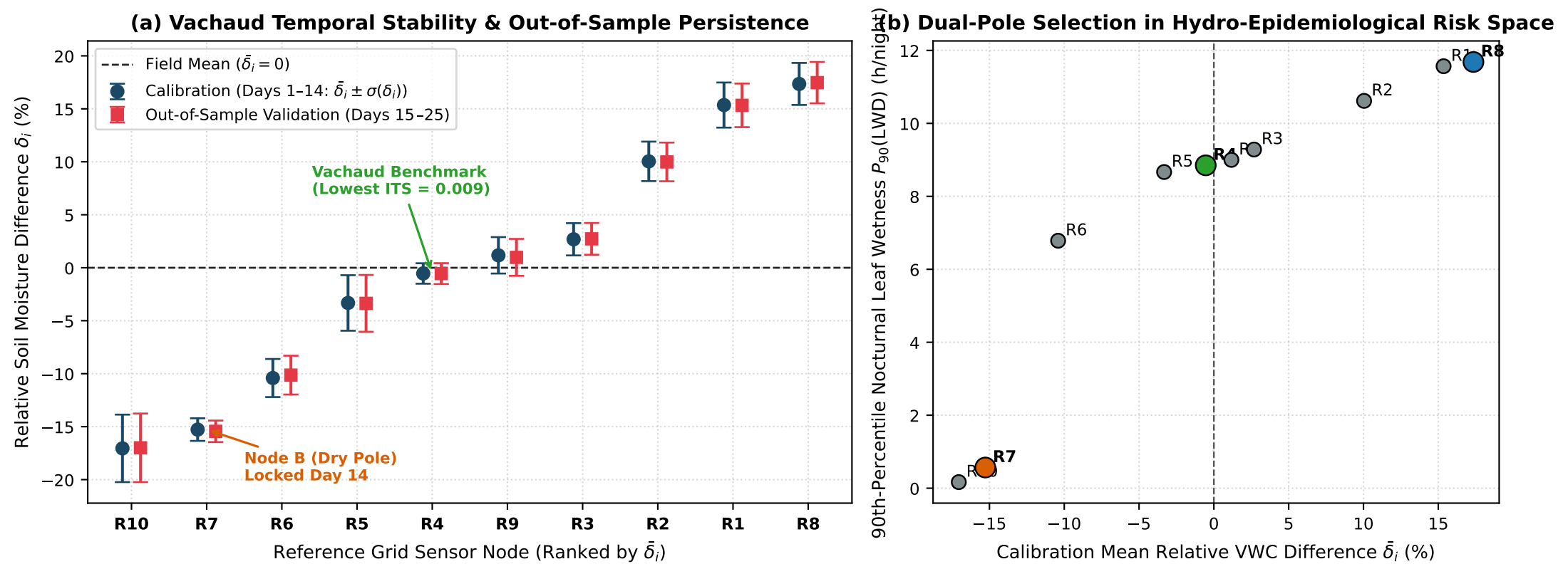
(a) Vachaud Temporal Stability & Out-of-Sample Persistence
Field Mean ($\bar{\delta}_i = 0$)
Calibration (Days 1–14: $\bar{\delta}_i \pm \sigma(\delta_i)$)
Out-of-Sample Validation (Days 15–25)
Vachaud Benchmark
(Lowest ITS = 0.009)
Node B (Dry Pole)
Locked Day 14
Relative Soil Moisture Difference δ_i (%)
Reference Grid Sensor Node (Ranked by $\bar{\delta}_i$)
R10
R7
R6
R5
R4
R9
R3
R2
R1
R8
(b) Dual-Pole Selection in Hydro-Epidemiological Risk Space
90th-Percentile Nocturnal Leaf Wetness P_{90}(LWD) (h/night)
Calibration Mean Relative VWC Difference $\bar{\delta}_i$ (%)